

Manage the patient with an unsuppressed viral load (VL ≥ 50)

Assess and manage possible causes of unsuppressed viral load (VL ≥ 50):

- If pregnant or breastfeeding → 166.
- Check for underlying causes of unsuppressed VL, especially adherence issues and medication interactions → 173.
- Emphasise condom use and contraception, especially while VL is unsuppressed

- If patient is *not* on TLD (or ALD), check if same day ART switch is appropriate → 117.
- If patient is on DTG-based regimen, continue below.

Repeat VL in 3 months:

Second viral load result < 50

Continue routine VL monitoring → 112.

Second viral load result ≥ 50

- Increase efforts to resolve adherence¹ issues and address possible drug-drug interactions → 173.
- Manage further according to duration:

Patient has been on DTG-based regimen for less than 2 years

No

Adherence considered poor.

- No resistance testing needed. If drug interactions suspected, then discuss with HIV expert or HIV hotline → 178.
- Continue to address adherence and possible interactions.
- Repeat VL at next scheduled routine VL.

Patient has been on DTG-based regimen for at least 2 years.

- Assess adherence in last 6-12 months by checking script for pharmacy refills and notes for clinic appointment attendance².
- Have refills been collected > 80%³ of time or has patient attended > 80%⁴ clinic visits?

Yes

Adherence considered good.

Has patient had 2 or more VL results ≥ 1000 after starting DTG-based regimen?

No

Has patient had at least one VL ≥ 1000 with either: CD4 < 200, or an opportunistic infection⁵?

No

Has patient failed previous regimen before s/he started DTG-based regimen (on TLD2 or ALD2/2nd-line)?

No

- Discuss need for resistance testing and choice of new individualised regimen with HIV expert, infectious disease specialist, third line ART committee or HIV hotline → 178.
- If VL ≥ 1000, monitor CD4 6 monthly. If CD4 ≤ 200, restart co-trimoxazole → 113.
- Repeat VL 3 months after starting new regimen.

Yes

Virological failure confirmed.

¹Resistance to a DTG-based regimen is rare – the most probable cause for VL non-suppression is poor adherence. ²If available, also do drug level on urine or blood specimen: adherence is considered good if medications are detected in patient's urine/blood. ³Calculate adherence % for pharmacy refills: 'number of actual refills done during period assessed' ÷ 'number of months in period assessed'. Then x by 100. ⁴Calculate adherence % for clinic attendance: 'number of scheduled visits actually attended by patient during period assessed' ÷ 'number of scheduled visits during period assessed'. Then x by 100. ⁵Examples of opportunistic infections include TB, *Cryptococcal disease*, *Pneumocystis jirovecii* pneumonia (PJP), *Cryptosporidium*, *Isospora belli* (*Cystoisospora belli*).